

Cyclooxygenase Inhibition and the Spinal Release of Prostaglandin E₂ and Amino Acids Evoked by Paw Formalin Injection: A Microdialysis Study in Unanesthetized Rats

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Injection of formalin into the hind paw evokes a biphasic flinching of the injured paw. Pharmacological characterization of this behavior has implicated the spinal release of excitatory amino acids (EAAs) and cyclooxygenase (COX) products. To address this hypothesis, we examined the effect of paw formalin injection on release of EAAs and prostaglandin E₂-like immunoreactivity (PGE₂-LI) from the spinal cord in unanesthetized rats using a dialysis probe placed in the lumbar subarachnoid space. To assess the contribution of spinal COX products, the effects of *S*(+)- and *R*(-)-ibuprofen (active and inactive COX inhibitors) were examined. Paw formalin injection evoked a biphasic spinal release of PGE₂-LI with an increase above resting concentrations of 110% in the 0–10 min sample, and of 83% in the 20–30 min sample. Significantly increased release of glutamate (Glu; 110%) and aspartate (Asp; 112%) was only observed in the 0–10 min sample. Saline injection into the paw had no effect on behavior, PGE₂-LI, or EAA release. Intraperitoneal administration of 10 mg/kg, but not 1 mg/kg, *S*(+)-ibuprofen reduced paw flinching, blocked the elevated levels of PGE₂-LI, and suppressed Glu and Asp release to 50% of control. Intrathecal delivery of 10 µg, but not 1 µg, *S*(+)-ibuprofen also suppressed formalin-induced behavior, PGE₂-LI, Glu, and Asp release. *R*(-)-ibuprofen showed no effect on formalin-induced behaviors or spinal release. These data demonstrate that paw formalin injection produces spinal release of PGE₂-LI corresponding to the biphasic behavioral response and that the evoked release is blocked by antinociceptive doses of COX inhibitors. The suppressant effect of COX inhibition on EAA release indicates that prostaglandins may be involved in facilitation of afferent-evoked EAA release in the spinal dorsal horn.

[Key words: intrathecal microdialysis, formalin test, pain, antinociception, cyclooxygenase, prostaglandin E₂, excitatory amino acids, ibuprofen]

Protracted activation of small diameter primary afferents, as occurs following peripheral injury and inflammation, leads to a state of exaggerated spinal nociceptive processing that is be-

lieved to underlie the behavioral phenomena of hyperalgesia (see Woolf and Chong, 1993; Yaksh, 1993). Investigations of the pharmacology of this hyperalgesia have suggested an important and interlinking role for spinal excitatory amino acids (EAAs) and cyclooxygenase (COX) products (Malmberg and Yaksh, 1992a,b).

COX enzymes are found within neurons of the brain and spinal cord (Breder et al., 1993) and prostanoids are formed in the CNS, released from the brain (Navarro et al., 1989) and the spinal cord (Romero et al., 1984). Spinal cord release of prostanoids is evoked following C-fiber stimulation in frogs (Ramwell et al., 1966), in response to noxious thermal stimulation in rats (Coderre et al., 1990), after superfusion with high potassium solution (Yaksh, 1982), or during knee joint inflammation (Sorkin, 1992). The spinal delivery of prostanoids induce a state of hyperalgesia (Uda et al., 1990; Minami et al., 1992). Consistent with these observations, the spinal delivery of prostaglandin receptor antagonists (Malmberg et al., 1994) or COX inhibitors have minimal effect on nociceptive tests involving acute thermal or mechanical stimuli (Yaksh, 1982; Malmberg and Yaksh, 1992a), but are typically active in tests involving hyperalgesia (Yaksh, 1982; Taiwo and Levine, 1988; Malmberg and Yaksh, 1992b), or in models with a prolonged nociceptive stimulus, such as the formalin test (Huskaar et al., 1986; Chapman and Dickenson, 1992; Malmberg and Yaksh, 1992a, 1993a, 1994).

Current observations have suggested that many of these hyperalgesic states share the additional characteristic of being antagonized by spinally delivered glutamate (Glu) receptor antagonists (Coderre and Melzack, 1992; Yamamoto and Yaksh, 1992). Support that spinal Glu receptor activation may mediate such a hyperalgesic component is suggested by the ability of Glu receptor agonists to induce a thermal hyperalgesia (Malmberg and Yaksh, 1992b).

A well-characterized behavioral model that appears to incorporate several of the above variables is the response of an unanesthetized rat to a subcutaneous injection of a dilute formalin solution into the hind paw. The formalin injection induces a biphasic (phase 1, 0–10 min; phase 2, 20–60 min) flinching, shaking, and sometimes licking behavior of the injured paw (Wheeler-Aceto et al., 1990). The phase 1 behavior is believed to correspond to the high level of activity in the primary afferents that innervate the injured region, while phase 2 appears to result from the continuous low level of C-fiber input together with a facilitated state of spinal processing (Dickenson and Sullivan, 1987; Heapy et al., 1987; for discussion see Yaksh, 1993). Intrathecal administration of *S*(+)-ibuprofen, but not the inactive

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isomer *R*(-)-ibuprofen, suppressed the behavioral response evoked by formalin injection (Malmberg and Yaksh, 1992a). The stereospecific effect suggests that the antinociceptive effect of these agents was mediated by inhibition of prostaglandin synthesis and that prostaglandins may be one of the facilitating substances contributing to the behavioral responses in the formalin test. However, it has not yet been demonstrated that formalin injection into the paw induces prostaglandin release in the spinal cord concomitant with behavioral activity.

The aim of the present study was to characterize concurrently the time-dependent release of prostaglandin E_2 -like immunoreactivity (PGE₂-LI) and amino acids following the injection of formalin into the hind paw of the unanesthetized rat. If there is, indeed, a correspondence between spinal prostanoid release and behavior, the delivery of anti-hyperalgesic doses of a COX inhibitor should produce a correlated stereospecific [*S*(+)-ibuprofen > *R*(-)-ibuprofen] reduction in flinching behavior and spinal prostaglandin release. Moreover, as prostanoids have been shown to augment release of substance P from primary afferents (Nicol et al., 1990), and given the cocontainment Glu with substance P (De Biasi and Rustioni, 1988), decreased extracellular prostanoids secondary to COX inhibition may similarly serve to decrease the formalin-evoked release of Glu.

Materials and Methods

Experiments were carried out according to protocols approved by the Institutional Animal Care and Use Committee of the University of California, San Diego.

Construction of microdialysis fiber. The dialysis tube was constructed from a 19 cm Cupraphan hollow fiber with an inner diameter of 200 μ m, an outer diameter of 300 μ m, and 45 kDa molecular weight cut off (Filtral, AN 69-HF). The fiber was coated with a thin layer of epoxy (Devcon Corporation, Danvers, MA), along the whole length, except for a 4 cm region in the middle. In order to make the fiber firm enough for implantation, a Nichrome-Formvar wire with a 0.0026 inch inner diameter (A-M systems, Inc., Everett, WA), was passed through the fiber. Both ends of the fiber was then attached to polyethylene catheters (PE-10) with cyanoacrylate (Borden, Inc., Columbus, OH). A "U-shaped" loop was formed by gently bending the whole fiber in the middle, and the two pieces were bound together at the fiber-PE-10 connection using epoxy.

Animal preparation. Male Sprague-Dawley rats (320–350 gm, Harlan Industries, Indianapolis, IN) were implanted with the dialysis "loop" under halothane anesthesia, according to the method described by Marsala et al. (1994). The dialysis probe was inserted through an incision in the cisternal membrane and extended to the rostral edge of the lumbar enlargement. The two ends of the dialysis probe, consisting of PE-10 catheters, were externalized on the top of the head. After the implantation, the dialysis catheter was flushed with artificial cerebrospinal fluid (ACSF; see below). For intrathecal injection studies, a stretched PE-10 catheter (8.5 cm) was implanted to allow drug delivery to the lumbar level of the spinal cord. The dialysis experiments were performed 3 to 7 d after the initial surgery. Rats that showed any neurological deficits after surgery were immediately euthanized. The rats were only used once for dialysis experiments.

Dialysis and sample collection. For the experiment, one of the externalized PE-10 tubings was connected to a syringe pump using a length of PE-50 tubing (inflow). The other was attached to approximately a 20–30 cm PE-10 catheter that served as outflow. The dialysis system was perfused with ACSF consisting of 151.1 mM Na⁺, 2.6 mM K⁺, 0.9 mM Mg²⁺, 1.3 mM Ca²⁺, 122.7 mM Cl⁻, 21.0 mM HCO₃⁻, 2.5 mM HPO₄⁻, and 3.5 mM dextrose. The ACSF was bubbled with 5% CO₂ in 95% O₂ to adjust pH to 7.2. The ACSF was preheated to 37°C prior to the experiment. The flow rate was 10 μ l/min. Samples were collected in 10 min intervals in polypropylene tubes on ice and frozen at -20°C until analysis of PGE₂-like immunoreactivity (PGE₂-LI) by radioimmunoassay (RIA) and amino acids by high-performance liquid chromatography (HPLC) (see below). For each 10 min sample (100 μ l), 50 μ l of dialysate was used for determination of PGE₂-LI and 25 μ l for amino acid analysis.

In vitro/in vivo recovery of PGE₂-LI. In vitro recovery of PGE₂ through the dialysis membrane was determined by placing the dialysis probe in 37°C ACSF containing known concentrations of PGE₂. The probe was then perfused with ACSF at a flow rate of 10 μ l/min. Sampling was started after a 30 min perfusion. The PGE₂-containing solution was also assayed and the recovery was calculated as the ratio between the concentration in the sampled perfusates and the outside media. Three catheters were used for these studies, and two perfusion samples were collected per catheter.

To estimate *in vivo* recovery, cisternal CSF samples were collected and assayed for PGE₂-LI. Samples (10 μ l) of CSF from the cisterna magna was aspirated with a Hamilton syringe in halothane (1%) anesthetized rats.

Nociceptive testing: the formalin test. After a washout period of 30–40 min followed by 20 min of sample collection for baseline, the rats were gently restrained and 50 μ l of 5% formalin solution was injected subcutaneously into the dorsal surface of the right hind paw with a 30 gauge needle. The rats were then returned to a Plexiglas chamber for observation of the paw injected with formalin. A mirror was positioned on the opposite side of the Plexiglas chamber to allow unhindered observation. Formalin injection produced a characteristic flinching and occasional licking of the injected paw, without appearance of other behaviors. Pain-related behavior was, therefore, quantified by counting the incidence of flinching of the injected paw. The number of flinches were noted for every min during the first 6 min after formalin injection. Thereafter, starting at 10 min after formalin injection, the incidence of paw flinching was counted at 5 min intervals in periods of 1 min. The total observation period was 60 min. After the observation period, animals were sacrificed with an overdose of barbiturate mixture (Beuthanasia, 50 mg/kg, intraperitoneally, Schering-Plough Animal Health Corp., Kenilworth, NJ).

Chemical analysis. Concentration of PGE₂-LI in the perfusion samples was quantified using commercially available PGE₂[¹²⁵I] radioimmunoassay kit (Advanced Magnetics Inc., Cambridge, MA) in accordance with the manufacturer's protocol. A standard curve was constructed between 10 and 2800 fmol of PGE₂ per assay tube for analysis of the perfusate samples. Cross-reactivity of the antibody for other prostaglandins and metabolites was reported in the manufacturer's brochure to be 50% for PGE₁, 6% for PGA₁, 1.9% for PGA₂, less than 1.8% for PGB₁, PGB₂, 6-keto-PGE₁, PGF₁, PGF₂, and less than 1% for 6-keto-PGF₁, PGD₂, dihydro-keto-PGE₂, dihydro-keto-PGF₂, arachidonic acid, tromboxane B₂, 15-keto-PGE₂, 15-keto-PGF₂, 5-hydroxyarachidonic acid (HETE), 12-HETE, and 15-HETE measured at the 50% bound/free point. In the present study, it is unlikely that the immunoreactivity detected represent PGE₁ to any significant extent, since the PGE₂ precursor arachidonic acid is far more abundant in CNS of mammals than the PGE₁ precursor eicosatrienoic acid. Moreover, PGE₁ has not been detected in brain tissue (Steinhauer et al., 1979).

The amino acids Glu, aspartate (Asp), taurine (Tau), glycine (Gly), Serine (Ser), asparagine (Asn), glutamine (Gln), and citrulline (Cit) were analyzed by a phenyl isothiocyanate precolumn derivatization method (Cohen and Strydom, 1988), using a Waters HPLC with a reverse phase C18 and a UV detector. Sensitivity was 5–10 pmol/injection. Methionine sulfone was added to each sample as an internal standard. External standards, containing 40, 400, and 4000 pmol of authentic amino acids, were run at the beginning and end of each sample group. The amino acid peak heights were initially normalized to the methionine sulfone peak and then quantified based on a linear relationship between peak height and amounts of corresponding standards.

Drugs and injections. *S*(+)- and *R*(-)-ibuprofen was provided by the Ethyl Corporation (courtesy of D. P. Bauer and C. W. Matthews, Ethyl Corporation, Baton Rouge, LA) and dissolved in a solution of 5% 2-hydroxy-propyl- β -cyclodextrin (RBI; Research Biochemicals Inc., Natick, MA). For systemic administration, the agents were mixed such that a 300 gm rat received an injection of 300 μ l of the drug. For intrathecal injection studies, agents were mixed such that all doses were delivered in a total volume of 10 μ l followed by 10 μ l of saline to flush the catheter. Drug treatment or control injections of the vehicle, 5% cyclodextrin, was also performed 25 min before the injection of formalin into the paw. In addition, separate groups of animals were used to study the effect of the highest doses of intrathecal or intraperitoneal *S*(+)-ibuprofen on basal concentrations of PGE₂-LI. The treatment was performed after a 30 min washout period, and 10 min fractions was collected for a period of 60 min. These rats did not receive formalin injection.

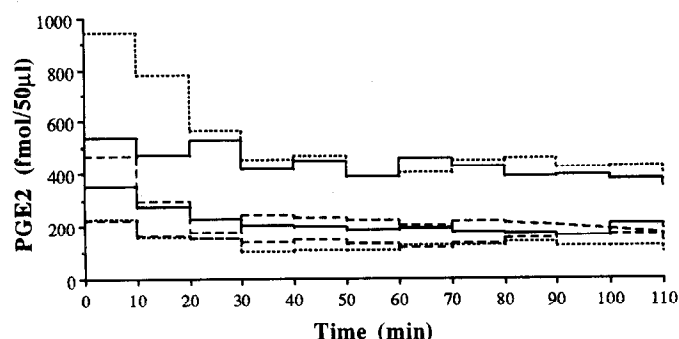


Figure 1. Spinal concentrations of PGE₂-LI in six individual rats from the start of infusion to 110 min after the initiation of perfusion. The perfusion flow rate was 10 µl/min. Equilibrium was obtained after approximately 30 min perfusion.

Data analysis and statistics. The time-response data of the behavior evoked by the formalin test are presented as the mean number of flinches $\pm$ SEM per minute for every minute during the first 6 min and thereafter at 5 min intervals for 60 min. Statistical analysis of the behavior data was accomplished by one-way ANOVA test followed by Student–Newman–Keuls multiple comparisons test provided that the *F* ratio gave $p < 0.05$. The release data for PGE₂ or amino acids within the treatment group were analyzed using repeated measures ANOVA. Comparison of release (percentage change from resting release) evoked by formalin or saline injection into the paw was accomplished using Mann–Whitney test. Treatment-group comparisons were made using the Kruskal–Wallis test followed by Dunn’s test for multiple comparisons.

Results

General behavior and motor effects

All rats showed normal behavioral and motor activity immediately after implantation of the dialysis probe and on the day of experimentation (3–7 d after implantation). None of the rats showed any behavioral changes during the ACSF perfusion.

Resting release conditions

The first 30 min after initiation of ACSF perfusion the levels of PGE₂-LI decreased, followed by a stable level of basal release for at least a 90 min period (Fig. 1). As indicated, though there were variances in the absolute levels between animals, the profile and time course was similar in all animals. Amino acid concentrations showed similar resting release profiles, with a stable baseline being attained some 20–30 min after initiation of the perfusion (data not shown). Baseline concentrations were defined as the mean of two consecutive 10 min samples taken after the initial 30–40 min washout period (Table 1). The lower number of animals in the amino acid groups, compared to the PGE₂ group, is due to an extra group of animals that was used to study the effect of *S*(+)-ibuprofen on resting release concentrations of PGE₂ without performing the formalin test. This group was not assayed for amino acids.

In vitro recovery experiments indicated that perfusion at a flow rate of 10 µl/min showed a steady-state dialysate concentration of $32 \pm 4\%$ of the external PGE₂ concentration. Collection of cisternal CSF ($n = 6$) and lumbar dialysates showed a concentration of PGE₂-LI of 110 pmol/ml, indicating that the *in vivo* recovery was approximately 10%, if cisternal and lumbar PGE₂ levels are similar.

Effect of saline or formalin injection into the paw

After the collection period to obtain baseline concentrations, 50 µl of 5% formalin was injected into the dorsal surface of the

Table 1. Resting release of PGE₂ and amino acids using intrathecal microdialysis

	N ^a	Mean $\pm$ SEM ^b	Median ^b
Resting release (pmol/ml)			
PGE ₂ -LI	65	10.0 $\pm$ 0.8	8.4
Resting release (nmol/ml)			
Glu	48	4.8 $\pm$ 0.4	4.4
Asp	48	0.8 $\pm$ 0.1	0.7
Tau	48	9.2 $\pm$ 0.4	8.8
Cit	44 ^c	4.4 $\pm$ 0.4	4.0
Gly	48	23.6 $\pm$ 1.2	21.2
Ser	48	10.4 $\pm$ 1.2	19.2
Asn	48	4.8 $\pm$ 0.4	4.0
Gln	48	60.0 $\pm$ 2.4	60.0

^a Number of animals.

^b Mean or median of the mean dialysate concentration in two consecutive 10 min samples collected after 30–40 min of washout.

^c In 4 of the 48 samples, Cit was not detected.

hindpaw. The formalin injection produced a characteristic biphasic flinching behavior of the injected paw, in contrast to paw saline injection, which induced minimal behavioral effects (Fig. 2). Comparisons within the groups (repeated measures ANOVA) indicated significant changes of PGE₂-LI and EAA release following formalin injection, but not after saline injection. Comparing the percentage changes evoked by formalin injection to the effect produced by saline injections during the first 10 min (phase 1) showed a significant increase (Mann–Whitney test) in spinal levels of PGE₂-LI, Glu, Asp, Tau, and Cit but not Gly, Ser, Asn, and Gln following formalin injection (Table 2). During the second phase flinching period evoked by formalin injection, only the concentrations of PGE₂-LI were significantly elevated compared to saline injection (Fig. 2). This occurred in the time interval 20–30 min after formalin injection. In the figures, the data are presented as percentage change from baseline values measured before the formalin injection, because the percentage changes were similar between rats and independent of absolute resting release levels, which varied between rats. The time course for saline and formalin-evoked flinching behavior, PGE₂, and Glu release is illustrated in Figure 2.

Effect of systemic or spinal *S*(+)-ibuprofen

Intraperitoneal injection of 10 mg/kg, but not 1 mg/kg, *S*(+)-ibuprofen given 25 min before formalin injection, produced a small, but significant suppression ($p < 0.05$, ANOVA) of phase 1 flinching activity and a more substantial reduction ($p < 0.001$; ANOVA) of the formalin-evoked flinching behavior during phase 2 (10–60 min) (Fig. 3). The effect was most prominent during the first part of the second phase, which corresponds to the significant increase in concentrations of PGE₂-LI during this time interval (Fig. 2). We, therefore, calculated the effects of *S*(+)-ibuprofen on flinching activity and PGE₂ release on the first part of phase 2; 20–30 min after formalin injection (Table 3). Similar to the behavioral effects, the formalin-evoked release of PGE₂-LI was suppressed by 10 mg/kg, but not 1 mg/kg, *S*(+)-ibuprofen (Table 3). Injection of 10 mg/kg of *S*(+)-ibuprofen to rats without formalin injection showed no significant effect ($p > 0.05$; repeated measures ANOVA) on resting release concentrations of PGE₂-LI, although the PGE₂-LI levels were decreased by approximately 20% ($n = 4$) lower than basal release, in the

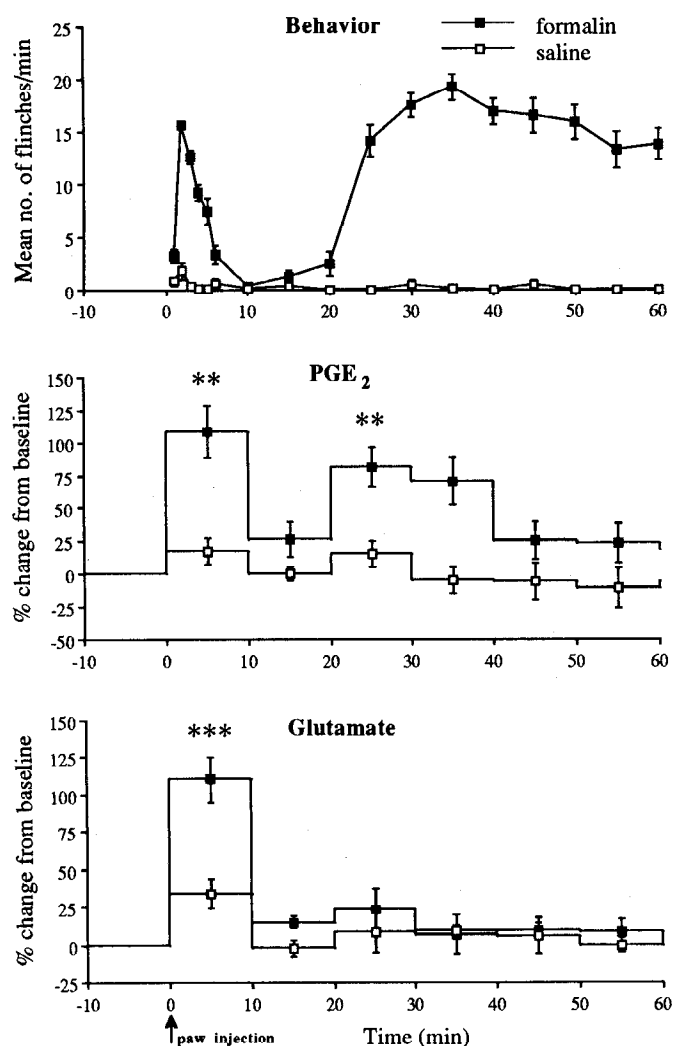


Figure 2. Time-effect curve for flinching behavior (*top*), and spinal release of PGE₂-LI (*middle*) or Glu (*bottom*), following saline or formalin injection into the rat paw. The behavioral data are presented as the mean $\pm$ SEM number of flinches versus the time after paw injection. Spinal release of PGE₂-LI or Glu is presented as the mean $\pm$ SEM percentage change from baseline versus time after formalin or saline injection. The asterisks indicate significantly increased release concentrations evoked by formalin compared to saline injection (Mann-Whitney test: **, $p < 0.01$; ***, $p < 0.001$).

interval 10–60 min following systemic *S*(+)-ibuprofen. Systemic *S*(+)-ibuprofen 10 mg/kg, but not 1 mg/kg, suppressed the formalin-evoked Glu, Asp, and Tau release by approximately 50% (Fig. 3, Table 4). Cit was even more reduced, by about 80% (Table 4). No effect was found on basal release of the amino acids. Intraperitoneal injection of the vehicle, 5% cyclodextrin ($n = 3$), produced no effect on basal concentrations or formalin-evoked elevations of PGE₂-LI or EAAs (data not shown).

Intrathecal administration of 10 μ g, but not 1 μ g, *S*(+)-ibuprofen, 25 min before formalin injection, significantly suppressed both phases of the formalin-evoked flinching behavior and blocked PGE₂ release (Fig. 4, Table 3). There was no effect of 10 μ g *S*(+)-ibuprofen on resting release ($p > 0.05$; repeated measures ANOVA; $n = 4$), and the PGE₂ concentration in the dialysates were only reduced by $5 \pm 7\%$ in the interval 30–40 min after the intrathecal injection. *S*(+)-ibuprofen (10 μ g) also

Table 2. Comparison of changes in spinal PGE₂ or amino acid concentrations following formalin or saline injection into the rat paw

	Phase	Time period	Percentage change from baseline levels $\pm$ SEM		
			Saline ($n = 6$)	Formalin ($n = 8$) ^a	Statistical significance ^b
PGE ₂	1	0–10 min	17 $\pm$ 15	109 $\pm$ 10	**
	2	20–30 min	15 $\pm$ 10	83 $\pm$ 15	**
Glu	1	0–10 min	34 $\pm$ 10	110 $\pm$ 15	***
Asp	1	0–10 min	27 $\pm$ 7	112 $\pm$ 20	***
Tau	1	0–10 min	32 $\pm$ 16	87 $\pm$ 14	*
Cit	1	0–10 min	8 $\pm$ 9	54 $\pm$ 14 ^a	*
Gly	1	0–10 min	13 $\pm$ 8	33 $\pm$ 9	NS
Ser	1	0–10 min	15 $\pm$ 10	42 $\pm$ 11	NS
Asp	1	0–10 min	19 $\pm$ 10	33 $\pm$ 10	NS
Gln	1	0–10 min	16 $\pm$ 5	24 $\pm$ 15	NS

^a In the formalin group, Cit was only detected in six animals.

^b The asterisks indicate significant difference between formalin and saline-injected rats using Mann-Whitney test: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

significantly suppressed (by approximately 50–60%) formalin-evoked release of Glu, Asp, Tau, and Cit, but had no effect on resting concentrations of these amino acids. The lower dose of *S*(+)-ibuprofen, 1 μ g, had no effect on formalin-induced amino acid release (Table 4). The vehicle, cyclodextrin at a concentration of 5% ($n = 2$), had no effect on either basal or evoked release of PGE₂-LI or amino acids (data not shown).

Effect of systemic or spinal *R*(-)-ibuprofen

R(-)-ibuprofen, administered intraperitoneally (10 mg/kg) or intrathecally (100 μ g), had no effect on the behavioral response evoked by paw formalin injection (Fig. 5, Table 3). Consistent with the behavior, neither systemic nor spinal delivery of *R*(-)-ibuprofen influenced the formalin-evoked PGE₂-LI and amino acid release (Fig. 5, Table 4).

Discussion

The correlation between behavior and release following noxious stimulation and additional covariance following pharmacological manipulation, gives important support for the involvement of a certain factor in mediating the behavioral effects. In the present study, we have focused on the spinal release of PGE₂-LI and amino acids. Spinal release of PGE₂ in rats has been previously measured using an *in vivo* perfusion technique (Coderre et al., 1990) and by spinal microdialysis (Sorkin, 1992; 1993b). Amino acids have also been measured previously, using a transverse microdialysis system through the spinal cord dorsal horn in anesthetized rats (Paleckova et al., 1992; Sorkin, 1993a,b) and in awake rats (Skilling et al., 1988; Sluka and Westlund, 1993). In the present study, the intrathecal loop-dialysis system permitted the experiments to be performed on awake rats up to several days after implantation and thereby minimize interference of acute surgical stress and anesthetics (Marsala et al., 1994). Moreover, the dialysis system, because of its small size, was compatible with concurrent placement of an intrathecal catheter allowing investigation of the effects of intrathecally administered COX inhibitors on spinal transmitter release as well as ongoing behavior.

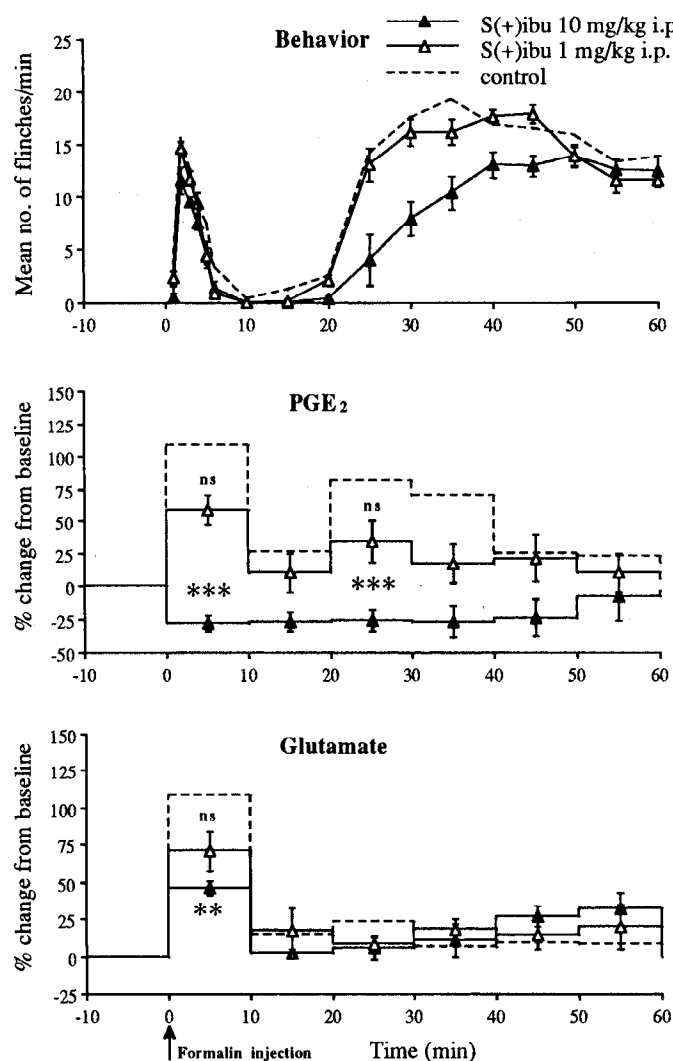


Figure 3. Time-effect curve for formalin-evoked behavior (*top*), and spinal release of PGE₂-LI (*middle*) or Glu (*bottom*), following intraperitoneal (*i.p.*) S(+)-ibuprofen treatment 25 min before the formalin test. The behavioral data are presented as the mean $\pm$ SEM number of flinches versus the time after formalin injection. Spinal release of PGE₂-LI or Glu are presented as mean $\pm$ SEM of the percentage change from baseline release versus time after formalin injection. The asterisks indicate that 10 mg/kg, but not 1 mg/kg (NS, $p > 0.05$), S(+)-ibuprofen produced a significant suppression of formalin-evoked release (Kruskal-Wallis followed by Dunn's multiple comparisons test: **, $p < 0.01$; ***, $p < 0.001$).

Recovery and resting release

In vitro experiments indicated that the PGE₂ recovery of the microdialysis probe was approximately 30%. However, this is likely to represent an optimistic situation compared to *in vivo* measurements, as the dialysis membrane is only surrounded by fluid, while in the subarachnoid space, parts of the dialysis fiber may be in contact with spinal cord tissue, thus decreasing the recovery. *In vivo* estimates of recovery for PGE₂ in the lumbar space was calculated on the basis of CSF sampled from the cisterna at the time of dialysis. The cisternal samples showed a PGE₂-LI concentration of approximately 100 pmol/ml, which was 10 times higher than perfusate concentrations, thus suggesting an *in vivo* recovery of approximately 10%. In most studies to date, CNS concentrations of prostanoids in rats have been measured in brain homogenates. The levels are in the order of

200 to 300 pmol/g rat brain tissue (Abdel-Halim et al., 1977; Abdel-Halim and Ånggård, 1979). Measurements of PGE₂ concentrations in the CSF of cats, dogs, and humans (Romero et al., 1984) showed lower levels in the CSF compared to the present study. This can be due to species differences or the fact that a catheter was implanted in the rats that previously been noted to produce higher PGE₂ concentrations compared to acute sampling (Romero et al., 1984). While the absolute levels varied between rats, resting levels of PGE₂, and amino acids in the dialysate did not change in any given rat (present study; Marsala et al., 1994). The reason for the differences in resting levels between animals is not clear. However, the exact orientation of the loop (dorsal or lateral surfaces of the cord) can vary between rats, and this may account for the interanimal baseline variation. This has not been systematically examined.

Afferent-evoked spinal amino acid release

Subcutaneous formalin injection led to a stimulus-dependent increase in the levels of Glu and Asp. The absolute concentrations of Glu were approximately 10 times the levels of Asp. This is in agreement with previous studies of noxious-evoked EAA release in the spinal cord (Skilling et al., 1988; Smullin et al., 1990; Paleckova et al., 1992). The present studies demonstrated that Glu and Asp concentrations were significantly increased only during the first 10 min period after the formalin injection. Previous studies have shown spinal Glu and Asp release after formalin, but the extended sampling time of 50 min precluded any specific association with a phase 1 or phase 2 response (Skilling et al., 1988). This early release probably corresponds to the intense activity in primary afferents following paw formalin injection (Heapy et al., 1987). Spinal delivery of antagonists for the Glu receptor subtype N-methyl-D-aspartate (NMDA) prior to the injection of formalin, but not between phase 1 and phase 2, reduce the second phase flinching behavior (Coderre and Melzack, 1992; Yamamoto and Yaksh, 1992). Such observations are consistent with the electrophysiological observations that spinal NMDA receptors play an essential role in the Glu release during the first phase of the formalin test. The levels of spinal Glu and Asp release were also slightly increased, to approximately 30–40%, during the second phase of the formalin test. This change, though not statistically significant, may be of importance. The second-phase behavior evoked by formalin injection is believed to be a result of continuous low level primary afferent activity and a facilitation of afferent sensory processing that is triggered by the initial burst of afferent input (see Dickenson and Sullivan, 1987). It is, thus, possible that spinal sensitization processes are resulting in an increased responsiveness of spinal neurons, so that less Glu and Asp are necessary to mediate similar behavioral responses as in phase 1 of the formalin test.

Formalin injection also produced elevation of Tau and Cit concentrations during the first phase. Tau has been suggested to be an inhibitory modulator during excitation (Lehmann et al., 1984). In agreement with this, substance P has been shown to evoke Tau release and low concentrations of Tau reduce substance P-induced behaviors (Smullin et al., 1990). Sciatic nerve stimulation also produced significant Tau release in the spinal cord (Paleckova et al., 1992). However, a previous study of spinal release evoked by the formalin test showed no significant release of Tau (Skilling et al., 1988). Cit was measured as an indicator of nitric oxide (NO) production, because Cit is formed in stoichiometric amounts with NO (Bredt and Snyder, 1989).

Table 3. Effect of intraperitoneal or intrathecal delivery of *S*(+)- or *R*(-)-ibuprofen on nociceptive behavior and spinal PGE₂ release evoked by paw formalin injection

Treatment	N	Formalin-evoked PGE ₂ release and behavioral response			
		Phase 1 (0–10 min)		Phase 2* (20–30 min)	
		Percentage change ^a	Flinches ^b	Percentage change ^a	Flinches ^b
Control (formalin)	8	109 ± 50	52 ± 4	83 ± 15	34 ± 4
<i>S</i> (+)ibu, 10 mg/kg (i.p.)	6	-28 ± 6**	35 ± 4*	-26 ± 8**	12 ± 4***
<i>S</i> (+)ibu, 1 mg/kg (i.p.)	6	60 ± 12 ^{NS}	43 ± 3 ^{NS}	34 ± 20 ^{NS}	31 ± 3 ^{NS}
<i>S</i> (+)ibu, 10 µg (i.t.)	6	7 ± 16**	35 ± 4*	-10 ± 10**	17 ± 2***
<i>S</i> (+)ibu, 1 µg (i.t.)	6	85 ± 16 ^{NS}	57 ± 3 ^{NS}	100 ± 20 ^{NS}	33 ± 2 ^{NS}
<i>R</i> (-)ibu, 10 mg/kg (i.p.)	6	93 ± 20 ^{NS}	44 ± 7 ^{NS}	70 ± 26 ^{NS}	26 ± 7 ^{NS}
<i>R</i> (-)ibu, 100 µg (i.t.)	6	125 ± 17 ^{NS}	44 ± 5 ^{NS}	79 ± 20 ^{NS}	32 ± 2 ^{NS}

^a Data represent the percentage change from baseline ± SEM. The asterisks that follow indicate if the treatment group is significantly different from the control group (Kruskal–Wallis followed by Dunn's multiple comparisons test: **, $p < 0.01$).

^b The mean ± SEM of the sum of paw flinches in the interval 20–30 min after formalin injection. The asterisks indicate significantly different numbers of flinches compared to the control group (ANOVA followed by Student–Newman–Keuls multiple comparisons test: *, $p < 0.05$; ***, $p < 0.001$).

NO synthesis inhibitors have been shown to suppress evoked Cit release (Sorkin, 1993a) and produce antinociception in the formalin test (Malmberg and Yaksh, 1993b), which is consistent with elevated levels of Cit following formalin injection. The inhibitory neurotransmitter Gly appears not to be involved in the formalin test. This is in agreement with the studies by Skilling and colleagues (Skilling et al., 1988). Ser, Asn, and Gln are believed to have predominantly metabolic roles.

Afferent-evoked spinal PGE₂ release

Formalin injection into the paw evoked a biphasic increase in the extracellular levels of PGE₂-LI with a time course that paralleled the phase 1 and phase 2 behaviors and the firing pattern of wide dynamic range neurons examined following the formalin test (Dickenson and Sullivan, 1987). The formalin-evoked Glu release may mediate the elevated PGE₂ levels. This is supported by studies showing that spinal delivery of Glu receptor agonists mediate a thermal hyperalgesia that is attenuated by intrathecal COX inhibitors (Malmberg and Yaksh, 1992b). Moreover, spinal administration of NMDA has been demonstrated to evoke PGE₂ release *in vivo* (Sorkin, 1993b), and in spinal cord slices *in vitro* (Malmberg and Yaksh, unpublished observations).

Effect of *S*(+)- and *R*(-)-ibuprofen on formalin-evoked release

In the present studies, using spinal or systemic doses of *S*(+)-ibuprofen that were shown to be effective in diminishing the flinching behavior in the formalin test, the behaviorally active, but not inactive dose, were able to produce a corresponding reduction in the formalin-evoked release of PGE₂-LI. That the low systemic dose of *S*(+)-ibuprofen (1 mg/kg) was ineffective, also supports the suggestion that intrathecal injection of *S*(+)-ibuprofen (10 µg) produced an antinociceptive effect by acting at the spinal cord level. The ability of the systemic dose to reduce spinal release of PGE₂-LI provides direct support that the systemically delivered agent can, indeed, exert a central action. This, of course, does not exclude any potential anti-inflammatory effects directed as well at the site of injury in the periphery. That the effects observed with both spinal and systemic *S*(+)-ibuprofen is mediated by COX inhibition is supported by the lack of effect of the inactive isomer, *R*(-)-ibuprofen. While *R*(-)-ibuprofen is converted to its active enantiomer *S*(+)-ibuprofen following systemic administration (Adams et al., 1976), it appears in these studies not to have contributed, perhaps because the systemic doses was inadequate.

Table 4. Effect of *S*(+)- or *R*(-)-ibuprofen on formalin-evoked release of Glu, Asp, Tau, and Cit during the first 10 min after formalin injection (phase 1)

Treatment	Percentage change from baseline and statistical significance ^a			
	Glu	Asp	Tau	Cit
Control (formalin)	110 ± 15	112 ± 20	87 ± 14	54 ± 14
<i>S</i> (+)ibu, 10 mg/kg (i.p.)	46 ± 5*	45 ± 9*	35 ± 8*	9 ± 6*
<i>S</i> (+)ibu, 1 mg/kg (i.p.)	71 ± 13 ^{NS}	98 ± 26 ^{NS}	61 ± 15 ^{NS}	35 ± 10 ^{NS}
<i>S</i> (+)ibu, 10 µg (i.t.)	50 ± 8*	52 ± 10*	35 ± 10*	15 ± 8*
<i>S</i> (+)ibu, 1 µg (i.t.)	91 ± 20 ^{NS}	99 ± 21 ^{NS}	66 ± 19 ^{NS}	45 ± 11 ^{NS}
<i>R</i> (-)ibu, 10 mg/kg (i.p.)	103 ± 29 ^{NS}	91 ± 33 ^{NS}	68 ± 11 ^{NS}	38 ± 14 ^{NS}
<i>R</i> (-)ibu, 100 µg (i.t.)	104 ± 37 ^{NS}	101 ± 18 ^{NS}	64 ± 13 ^{NS}	78 ± 15 ^{NS}

^a Data represent the mean percentage change from baseline ± SEM ($N = 6–8$; see Table 3). The asterisks that follow indicate if the treatment group is significantly different from the control group (Kruskal–Wallis followed by Dunn's multiple comparisons test: *, $p < 0.05$).

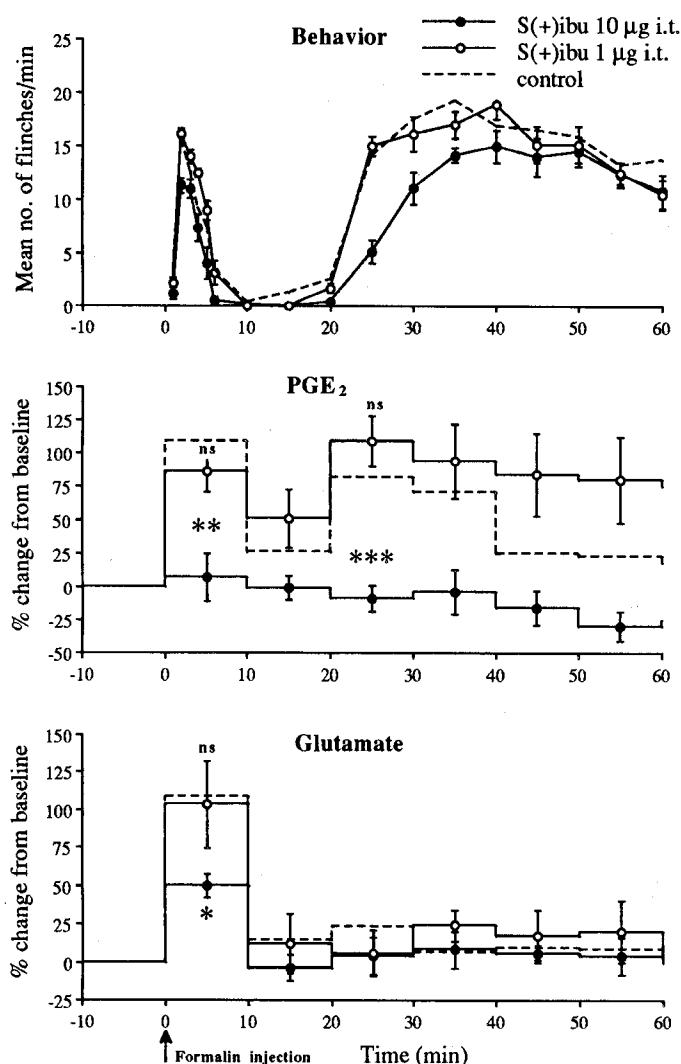


Figure 4. Time-effect curve for formalin-evoked behavior (*top*), and spinal release of PGE₂-LI (*middle*) or Glu (*bottom*), following intrathecal (*i.t.*) S(+)-ibuprofen 25 min before the formalin test. The behavioral data are presented as the mean $\pm$ SEM number of flinches versus the time after formalin injection. Spinal release of PGE₂-LI or Glu are presented as mean $\pm$ SEM of the percentage change from baseline release versus time after formalin injection. The asterisks indicate that 10 µg (*i.t.*) but not 1 µg (*NS*, $p > 0.05$), S(+)-ibuprofen produced a significant suppression of formalin-evoked release (Kruskal-Wallis followed by Dunn's multiple comparisons test: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$).

COX inhibition only produced a modest reduction of the phase 1 behavior, while more significant suppression was observed during phase 2 of the formalin test. In our previous study (Malmberg and Yaksh, 1992a), S(+)-ibuprofen produced a non-significant reduction of the phase 1 behaviors. The significant suppression of phase 1 in the present study may be due to the fact that the flinching activity was noted for every minute during the first 6 min after formalin injection. In our previous study, the first phase was defined as the sum of flinches during the periods 1–2 min and 5–6 min after the formalin test, which may underestimate a small reduction of phase 1. In any case, the effect of the COX inhibitors are more notable during the second phase. Evoked PGE₂ release during both phases 1 and 2 was totally blocked by COX inhibitors, while the behavior was reduced. This suggests that formalin-evoked elevation of spinal

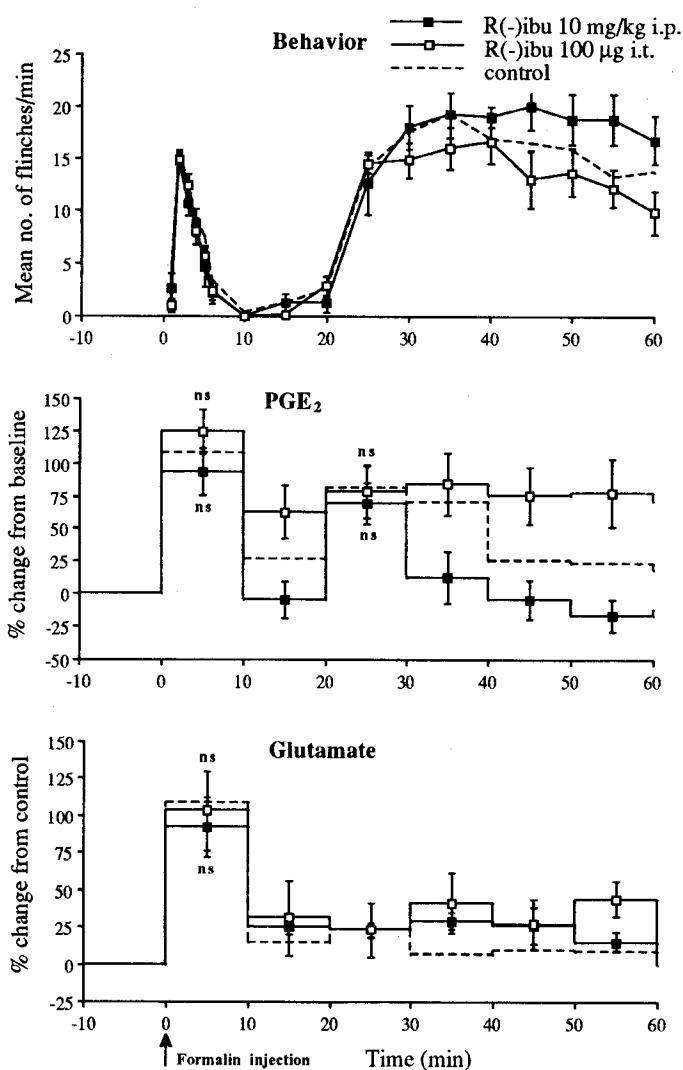


Figure 5. Time-effect curve for formalin-evoked behavior (*top*), and spinal release of PGE₂-LI (*middle*) or Glu (*bottom*), following intraperitoneal (*i.p.*) or intrathecal (*i.t.*) administration of R(-)-ibuprofen treatment 25 min before the formalin test. The behavioral data are presented as the mean $\pm$ SEM number of flinches versus the time after formalin injection. Spinal release of PGE₂-LI or Glu are presented as mean $\pm$ SEM of the percentage change from baseline release versus time after formalin injection. There was no significant effect of either intraperitoneal or intrathecal R(-)-ibuprofen on formalin-evoked release (Kruskal-Wallis followed by Dunn's multiple comparisons test: $p > 0.05$).

PGE₂ concentrations is only partly responsible for the behavioral profile in the formalin test. The greater effect of PGE₂ blockade on phase 2 flinching activity, compared to phase 1, suggest that the role of spinal PGE₂ is mainly associated with the facilitation of nociceptive processing, which is believed to occur during phase 2 of the formalin test. Although facilitatory substances, such as PGE₂, are also involved in the phase 1 response a greater portion of transmitters, such as Glu, are released during this phase probably due to the intense afferent barrage and are likely responsible for the phase 1 behavioral effect.

COX inhibition produced an approximately 50% reduction of formalin-evoked Glu release during the first phase of the formalin test. PGE₂ has been shown to enhance substance P release from sensory neurons (Nicol et al., 1992), and it is possible that PGE₂ also facilitates release of other neurotransmitters, such as

Glu and Asp. This idea is supported by observations that prostaglandins are involved in the evoked release of Asp in the brain (Dorman et al., 1986) and that PGE₂ facilitates afferent-evoked Glu and Asp release from the spinal cord using *in vivo* microdialysis (A. B. Malmberg, unpublished observation). In contrast to our findings, Sorkin showed no effect of ketorolac on Glu release evoked by NMDA activation (Sorkin, 1993b). It is possible that the discrepancy depends on different dialysis methods and nociceptive stimuli, but may also depend on the varying sensitivity of different inhibitors on COX enzymes (Mitchell et al., 1994). Interestingly, enhanced Asp release by prostaglandins was specifically suppressed by ibuprofen, but not indomethacin or acetylsalicylic acid (Dorman et al., 1986).

Finally, while the classical explanation for the action of COX inhibitors has focused on their peripheral action, the correlations between anti-hyperalgesic actions and their respective anti-inflammatory actions has been surprisingly poor (Brune et al., 1991; McCormack and Brune, 1991). The present studies showing a close correlation between dose and the respective effects on formalin/evoked behavior, and spinal prostanoid release adds further support for the hypothesis that these agents may exert a principle action by reducing the release of spinal prostanoids otherwise evoked by afferent input.

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